Zoo Cute AI Assistant, AR App, and Metaverse Companion

On-Chain Realization of the 2021 Vision (§§21–23)

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Abstract

The October 31, 2021 Zoo whitepaper described three interlocking experience surfaces: a personality-bearing "Cute AI Assistant" (§21), an Augmented Reality app for in-the-field encounters (§22), and a Metaverse Companion that follows the user across surfaces (§23). This paper documents the 2025-12-15 realization. The Cute AI Assistant ships as the on-device avatar within Zoo Desktop, backed by the ZEN model family co-developed with HANZO. The AR App ships as Zoo Field, a mobile companion tied to geo-attested quest completion. The Metaverse Companion ships as the persistent avatar binding wallet, NFT, chat, and AR surfaces, anchored by per-LLM chains [1]. We document architecture, dependencies, and the 2021 commitments each surface honours.

1 The Cute AI Assistant

1.1 2021 Commitment

§21 of the founder whitepaper described an AI assistant fused with each Zoo NFT: emotionally attuned, conversational, capable of recognizing user context and personalizing responses, and "ready to cheer you up." Each species would have a distinct personality profile.

1.2 2025-12-15 Realization

Ships as the on-device avatar within **Zoo Desktop** (Tauri + React, `~/work/zoo/app/`). Backed by the ZEN model family co-developed with HANZO AI: `eco-1`, `coder-1`, `nano-1` (see ZIP-0010). Personality is realized through:

- Per-species system prompt and persona file shipped with each NFT (deterministic; on-chain hash committed under ZIP-0204).
- Persistent semantic memory backed by the Experience Ledger (ZIP-0401, content-addressable semantic memory).
- Real-time conversational UX (ZIP-0417).
- Multimodal conservation knowledge (ZIP-0406).
- Local-first inference on the user's device (Hanzo Network GPU/CPU), with fallback to a Hanzo cluster when local resources are insufficient.

1.3 Personality Profiles (2021 Catalogue)

The eight V1 species each have a distinct, named personality articulated in the 2021 paper §21.1 and instantiated in 2025-12-15:

Species	**Persona (canonical, 2021)**
Red Wolf	Loyal pack-mate; encourages teamwork.
Amur Leopard	Reserved scholar; surfaces obscure facts.
Nubian Giraffe	Optimistic horizon-scout; long-term planner.
Sumatran Elephant	Memory-keeper; reminds the user of past interactions.
Pygmy Hippo	Shy comedian; light comfort and small talk.
Javan Rhinoceros	Gentle giant; encourages slow deep work.
Siberian Tiger	Focused mentor; pushes mastery and discipline.
Emperor Penguin	Stoic companion; weather-aware, calm.

Table 1: Canonical V1 personalities, instantiated in 2025-12-15 as prompt + behaviour files.

1.4 Architecture

- **Frontend.** Zoo Desktop (Tauri 2 + React 19), with the avatar rendered via the in-app 3D scene; voice synthesis local-only.
- **Inference.** Zen models served by Hanzo Network on a per-LLM chain (one chain per model, each model is a sovereign Quasar appchain — see [1]).
- **Memory.** Experience Ledger (content-addressable; semantic-vector indexed); BitDelta personalization (ZIP-0007); DeltaSoup community merging (ZIP-0009).
- **Identity.** Avatar is bound to the user's Zoo DID (ZIP-0033); on-device key custody.
- **Receipts.** Every inference call emits a TEE-attested receipt anchored on A-Chain (LP-134, Hanzo AI Chain).

1.5 Pre-ChatGPT Provenance

Notable historical fact: §21 of the October 31, 2021 whitepaper described a GUI-based conversational AI avatar *thirteen months before* CHATGPT's public launch (November 30, 2022). Zoo's avatar AI concept — including persistent semantic memory and personality binding — predates the mainstream LLM-chat paradigm. The Experience Ledger (`papers/zoo-experience-ledger/`) predates mainstream vector databases and CHATGPT's native conversation memory feature (February 2024).

2 Augmented Reality App (Zoo Field)

2.1 2021 Commitment

§22 described an AR app that lets the user encounter their Zoo NFT animal in real-world space, supporting in-the-field engagement and conservation education.

2.2 2025-12-15 Realization

Ships as **Zoo Field**, a mobile AR companion published under `~/work/zoo/apps/`. Anchored to:

- **Geo-attested quest completion.** On-site visits to partnered sanctuaries and conservation sites complete quests under ZIP-0034 (XP & Quests) with `ATTESTATION` verification.
- **Species-ID classification.** On-device species classification (ZIP-0510, Species Protection Monitoring) recognizes real animals and rewards the user for verified observations.
- **Habitat overlays.** Visualizes habitat-restoration data fed from ZIP-0505 (Reforestation Verification) and ZIP-0520 (Habitat Conservation).
- **Citizen-science contribution.** Submitted observations flow into the Citizen Science Contribution Protocol (ZIP-0602) and are eligible for impact-attestation.

2.3 Architecture

- **Stack.** Native iOS (ARKIT) and Android (ARCORE) with shared Rust core for protocol bindings.
- **Inference.** Local on-device for species classification and pose estimation; offloaded to Hanzo Network for harder cases.
- **Identity.** DID-bound (ZIP-0033); offline-capable with deferred attestation submission.
- **Privacy.** Geo data is selectively disclosed on Z-Chain (LP-134); raw GPS is never written publicly.

2.4 Field-Trip Mode

A guided mode for sanctuary visits (Foundation events). The visitor's DID accrues XP and unlocks limited-edition wildlife photography NFTs (ZIP-0205).

3 Metaverse Companion

3.1 2021 Commitment

§23 described a metaverse companion that follows the user across surfaces, persists state, and integrates with games (mini-games, Coin Runner, Memory Bridge — §24).

3.2 2025-12-15 Realization

The companion is the persistent avatar binding the user's Zoo DID (ZIP-0033) to:

- Wallet (Lux Wallet, white-label per LP-134).
- NFT portfolio (Zoo NFT Liquidity Protocol, `contracts/specs/zoo-nft-liquidity-protocol.md`).
- Chat (Zoo Desktop avatar, §1).
- AR (Zoo Field, §2).
- Virtual Habitat Simulation (ZIP-0300).
- Multiplayer Conservation Game (ZIP-0303).
- Spatial Web agents (ZIP-0011, ZIP-0433).

3.3 State Continuity

The companion's state (XP, KEEPER, attestations, semantic memory) is canonically held under the user's Zoo DID. Every surface reads from A-Chain attestation; writes back through Hanzo AI Chain receipts. Cross-surface continuity is therefore a function of identity, not a function of any single platform.

3.4 Game-Asset Interoperability

ZIP-0304 (Game Asset Interoperability) defines how items move across mini-games. ZIP-0210 (Cross-Collection Composability) defines how Zoo NFTs compose with non-Zoo collections through attested trait inheritance.

3.5 Why Holographic

The companion realizes the "holographic" axis named in the 2021 paper's 2025 update (§28): every sub-DAO, per-LLM chain, and regional chapter inherits the same governance topology and the same identity primitive. Whether the user is on Desktop, Field (AR), the DEX, or a sub-DAO sub-domain, the same DID drives the same companion. That is the engineering meaning of "holographic" in 2025-12-15 Zoo: a single template, replicated across every surface, anchored on the same chain.

3.6 Architecture Diagram (Logical)

```
Zoo DID      (ZIP-0033, A-Chain attested)
```

```
Companion  <------> Experience  (Ledger; ZIP-0401)
 State               Memory



 Surfaces
 - Zoo Desktop (avatar chat)
 - Zoo Field (AR)
 - Zoo Wallet
 - Zoo NFT Marketplace
 - Sub-DAOs (per-LLM chains)
```

4 Roadmap (Surface-Specific)

4.1 Shipped (as of 2025-12-15)

- Zoo Desktop avatar with Zen models, Experience Ledger memory, and DID binding.
- Per-LLM chain provisioning for the eight V1 personality profiles.
- Zoo Field AR v1 (iOS / Android) with species ID and geo-attested quest completion.

4.2 2025-12-25 Activation

- Zoo L1 mainnet cutover: companion state migrates from staging to mainnet.
- DID issuance opens for all existing 2021–2024 holders.
- ZIP-0034 Quest Registry seeded with the Founders' Quests.

4.3 Post-Mainnet (after 2025-12-25)

- Zoo Field AR v2: visual SLAM, multi-species scene, deeper habitat overlays.
- Multiplayer Conservation Game (ZIP-0303) public beta.
- Coin Runner mini-game (canonical 2021 design, §24.2.1).
- Memory Bridge mini-game (canonical 2021 design, §24.2.2).

4.4 Cross-References

- 2021 Whitepaper §21 (Cute AI), §22 (AR), §23 (Metaverse), §24 (Roadmap).
- Companion paper: *Zoo Per-LLM Chains.*
- Companion paper: *Zoo Experience Ledger.*
- Companion paper: *Zoo Spatial Web Agents.*
- Lux LP-132 (QuasarGPU), LP-134 (chain topology), LP-137 (GPU residency invariant).

References

[1] Zoo Labs Foundation, *Zoo Per-LLM Chains: One Sovereign Chain per Model*, 2025.